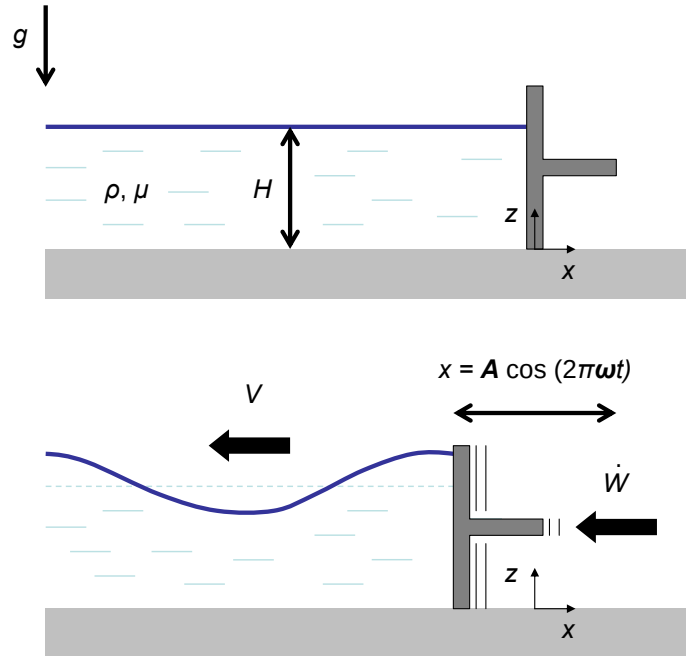


PROBLEM 3 (45 minutes)

The figure below illustrates a simple mechanism to generate periodic gravity waves in a pool of liquid (density ρ and viscosity μ), with undisturbed height H . The waves are driven by the harmonic motion of the lateral confining wall perpendicular to itself, with amplitude A and frequency ω .



1. (4 points). Use dimensional analysis to simplify the dependence of the power needed to drive the generator, $\dot{W}$, as a function of the governing parameters ($\rho, \mu, A, \omega, H, g$).
2. (1 point). Simplify the resulting dimensionless function for the case of negligible viscous forces. In the remainder of the problem, assume that this condition is satisfied.

To characterize the requirements of a new wave generator to be installed in a large water channel (subindex p), a scale model (subindex m) with $H_m/H_p = 1/20$ is tested in the lab using water as working fluid. Knowing that the wave frequency required in the water channel is $\omega_p = 1/6 \text{ s}^{-1}$,

3. (2 points). Determine the value of the frequency, ω_m , that must be used in the experiments to guarantee physical similarity.
4. (2 points). If the power used in the model is $\dot{W}_m = 1 \text{ Watt}$, obtain the power needed to drive the wave generator in the water channel, $\dot{W}_p$.
5. (1 point). Justify the hypothesis of negligible viscous forces. If needed, make a reasonable guess for the values of the involved variables.

PROBLEM 3

$$(1) \dot{W} = f(\bar{\rho}, \bar{\mu}, \bar{A}, \bar{\omega}, \bar{H}, \bar{g})$$

SINCE (ρ, A, ω) FORM AN INDEPENDENT SET, BUCKINGHAM'S Π -THEOREM SAYS

$$\Pi_{\dot{W}} = \Pi_{\dot{W}}(\Pi_{\mu}, \Pi_H, \Pi_g) \quad \text{WHERE} \quad \Pi_{\dot{W}} = \frac{\dot{W}}{\rho \omega^3 A^5} ; \Pi_{\mu} = \frac{\mu}{\rho \omega A^2}$$

$$\Pi_H = \frac{H}{A} ; \Pi_g = \frac{g}{\omega^2 A}$$

$$(2) \Pi_{\mu} \rightarrow 0 \Rightarrow \Pi_{\dot{W}} = \Pi_{\dot{W}}(\Pi_H, \Pi_g)$$

$$(3) \frac{H_p}{A_p} = \frac{H_m}{A_m} \Rightarrow \frac{A_p}{A_m} = \frac{H_p}{H_m}$$

$$\frac{g}{\omega_p^2 A_p} = \frac{g}{\omega_m^2 A_m} \Rightarrow \frac{\omega_m}{\omega_p} = \sqrt{\frac{A_p}{A_m}} = \sqrt{20} \Rightarrow \boxed{\omega_m = \frac{\sqrt{5}}{3} \text{ s}^{-1}}$$

$$(4) \left. \begin{array}{l} \Pi_{Hm} = \Pi_{Hp} \\ \Pi_{gm} = \Pi_{gp} \end{array} \right\} \Rightarrow \Pi_{\dot{W}m} = \Pi_{\dot{W}p} \quad \text{THUS} \quad \frac{\dot{W}_p}{\rho \omega_p^3 A_p^5} = \frac{\dot{W}_m}{\rho \omega_m^3 A_m^5} \Rightarrow$$

$$\Rightarrow \frac{\dot{W}_p}{\dot{W}_m} = \left(\frac{\omega_p}{\omega_m}\right)^3 \left(\frac{A_p}{A_m}\right)^5 = \left(\frac{A_p}{A_m}\right)^{7/2} = 20^{7/2} \Rightarrow \boxed{\dot{W}_p = 35.78 \text{ kWatt}}$$

$$(5) \Pi_{\mu} = \frac{\mu}{\rho \omega A^2} \ll 1 ? \quad \text{WATER} \left\{ \begin{array}{l} \rho = 10^3 \text{ kg m}^{-3} \\ \mu = 10^{-3} \text{ Pa s} \end{array} \right\} \Pi_{\mu} = \frac{10^{-6}}{\omega A^2}$$

$$\text{TAKING } A \sim 1 \text{ m} \\ \omega \sim 1 \text{ s}^{-1}$$

IT'S SEEN THAT Π_{μ} IS, INDEED VERY SMALL

(LARGE REYNOLDS NUMBERS)